

SIMATS ENGINEERING

ASSESSMENT TOOL 2: Scenario-Based Research Assessment

COURSE NAME: ARTIFICIAL INTELLIGENCE COURSE CODE: MLA01
AND EXPERT SYSTEMS

COURSE OUTCOME COVERED

CO1: *Apply search algorithms and heuristic techniques to solve state-space search and optimization problems. (BTL 3)*

Assessment Tool Weightage: 45 %

Total Marks: 50

Time: 60 Mins

No. of Questions: 5

Annexure A

Scenario-Based Research Assessment

Code of Conduct:

I Juhi Surin (192525148) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Juhi Surin

Criterion	Weightage	Marks Obtained
Identification of Latest AI Application	10%	
Problem Analysis and Domain Understanding	15%	
AI Technologies and Working Mechanism	20%	
Research Quality and Literature Review	10%	
Real-World Implementation and Case Analysis	10%	
Impact, Challenges, and Future Scope	15%	
Originality and Critical Thinking	10%	

Organization, Presentation, and Documentation	10%	
Total	100%	

Signature of the Faculty

Total Marks Awarded

ANSWER ANY ONE

No.	AI Domain	Scenario-Based Research Question
1	Healthcare	Healthcare organizations are increasingly adopting Artificial Intelligence to improve diagnosis, treatment, and patient monitoring. Identify and research a latest AI-based application in healthcare (such as AI-assisted medical diagnosis, AI radiology systems, robotic surgery, AI drug discovery, or virtual healthcare assistants). Analyze its working principles, AI techniques used, real-world implementation, benefits, and limitations.
2	Finance	Financial institutions are leveraging Artificial Intelligence to enhance security, decision-making, and customer services. Identify and research a latest AI application in the finance domain (such as AI fraud detection systems, AI-powered financial advisors, algorithmic trading platforms, or intelligent banking assistants). Discuss its functionality, technologies involved, advantages, and challenges.
3	Retail	Retail companies are adopting AI-driven solutions to provide personalized and intelligent shopping experiences. Identify and research a latest AI application in retail (such as Generative AI shopping assistants, AI recommendation engines, virtual try-on systems, or AI-based demand forecasting). Explain its application, AI methods used, impact on customers and businesses, and future scope.
4	Autonomous Vehicles	The transportation sector is rapidly evolving with AI-based autonomous systems. Identify and research a latest AI application in autonomous vehicles (such as self-driving technology, AI-based perception systems, autonomous delivery vehicles, or intelligent traffic management). Analyze the AI technologies involved, real-world deployment, benefits, and challenges.
5	Robotics	Intelligent robots are being developed for applications in industries, healthcare, education, and daily life. Identify and research a latest AI application in robotics (such as humanoid robots, collaborative robots, service robots, or AI-powered warehouse robots). Explain their capabilities, AI techniques, real-world usage, and future developments.
6	Manufacturing	Manufacturing industries are integrating AI to achieve smart and automated production environments. Identify and research a latest AI application in smart manufacturing (such as AI-based predictive maintenance, computer vision quality inspection, digital twins, or autonomous manufacturing systems). Discuss its working, industrial applications, advantages, and limitations.

Structure of the Report:

S.No	Report Component	Student Deliverable / Guiding Question
1	Title of AI Application	Provide the name of the latest AI application selected and the corresponding AI domain.
2	Domain Selection	Identify the application domain (Healthcare / Finance / Retail / Autonomous Vehicles / Robotics / Manufacturing).
3	Introduction and Background	Describe the AI application and provide an overview of its purpose and importance.
4	Problem Identification	Explain the real-world problem or challenge addressed by this AI application.
5	Need for AI Solution	Analyze why AI is required and how it improves traditional approaches.
6	AI Technologies Used	Identify the AI techniques involved (Machine Learning, Deep Learning, NLP, Computer Vision, Generative AI, Robotics, etc.).
7	Working Mechanism	Explain how the AI system works, including input data, processing methods, decision-making, and output generation.
8	Data Sources and Processing	Describe the type of data used and how the data is processed for intelligent decision-making.
9	Real-World Implementation	Identify organizations, industries, or real-world scenarios where this AI application is currently implemented.
10	Impact and Benefits	Evaluate the improvements achieved through the AI application in terms of efficiency, accuracy, cost, user experience, or automation.

11	Challenges and Limitations	Discuss technical, ethical, privacy, security, and deployment challenges associated with the application.
12	Future Scope and Emerging Trends	Explain possible enhancements, future developments, and research opportunities in this AI application.
13	Conclusion	Summarize the significance of the AI application and its contribution to solving real-world problems.
14	References	Provide research papers, technical articles, industry reports, and reliable sources referred for the study.

AI-Powered Radiology Triage and Diagnostic Imaging

How Foundation-Model Clinical AI is Reshaping Emergency Diagnosis

1. Title of AI Application

AI-Powered Radiology Triage and Diagnostic Imaging: a study of foundation-model clinical AI platforms, with Aidoc's CARE™ system as the primary case, alongside comparable tools such as Viz.ai and Qure.ai qXR. The corresponding AI domain is Healthcare.

2. Domain Selection

The application belongs to the Healthcare domain, specifically the sub-field of medical imaging and diagnostic radiology, where artificial intelligence is used to interpret CT scans, X-rays, and MRI studies, and to support clinicians in detecting and prioritizing life-threatening conditions.

3. Introduction and Background

Radiology has traditionally depended on the manual visual interpretation of medical images by trained radiologists. As imaging volumes have grown far faster than the radiology workforce, hospitals worldwide have faced longer reporting turnaround times and a heavier cognitive load on clinicians. Artificial intelligence, particularly deep learning applied to medical images, has emerged as a practical response to this gap.

Aidoc, founded in 2016, is presently the most widely deployed pure-play clinical radiology AI vendor. Its technology now runs across close to 2,000 hospitals worldwide and analyzes tens of millions of patient scans every year. Rather than a single-purpose algorithm, Aidoc's newer CARE™ platform is a foundation model, meaning one underlying model is trained broadly enough to recognize many different acute findings from a single CT scan, instead of needing a separate narrow model for every disease. Complementary tools such as Viz.ai (stroke detection) and Qure.ai's qXR (chest X-ray analysis) address the same underlying problem in adjacent clinical areas.

4. Problem Identification

The real-world problem is diagnostic bottleneck and delay in radiology reporting. Imaging demand has consistently outpaced the capacity of radiology departments to interpret scans quickly.

A Neiman Health Policy Institute study cited by Aidoc found that outpatient imaging interpretation turnaround times more than doubled between 2014 and 2023, with the sharpest increases in the most recent years. In emergency settings this delay is dangerous: a critical finding such as a brain hemorrhage, pulmonary embolism, or large-vessel stroke sitting unread in a queue can cost a patient

irreversible time. Reports note that delayed CT interpretation in emergency departments can add up to 150 minutes to a patient's stay, which directly affects clinical decision-making and patient throughput.

5. Need for AI Solution

Traditional radiology workflows process studies largely in a first-in-first-out order, which does not account for clinical urgency — a routine follow-up scan can sit ahead of a scan containing a life-threatening bleed simply because it was uploaded first. Human readers, however skilled, are also subject to fatigue, especially during long shifts or overnight coverage, and radiologist shortages mean many hospitals cannot simply add more staff to close the gap.

AI addresses this by continuously and tirelessly scanning every incoming study the moment it is captured, flagging suspected critical findings within minutes, and re-ordering the radiologist's worklist so the most urgent cases are opened first. This does not replace the radiologist's judgment; it changes the order and speed at which human judgment is applied, which is precisely where traditional approaches fall short.

6. AI Technologies Used

Core techniques

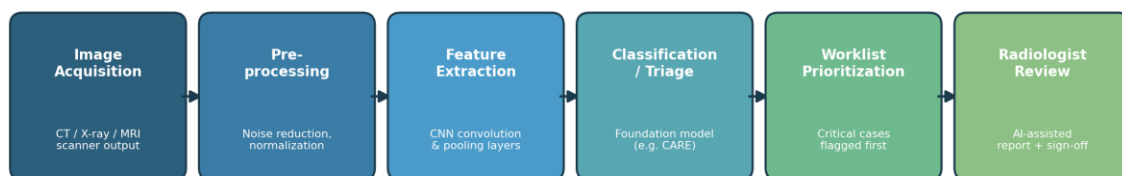
- Convolutional Neural Networks (CNNs) — the primary deep learning architecture used to learn visual patterns directly from raw pixel data in CT, X-ray, and MRI images, without needing hand-crafted features.
- Deep learning-based image segmentation — used to outline organs, vessels, and abnormal regions (e.g., cerebrovascular structures, lung nodules) for precise localization.
- Foundation models — a newer generation of large models (e.g., Aidoc's CARE™) trained across many indications simultaneously, and vision-language models (e.g., NVIDIA Clara Reason) that can output step-by-step reasoning alongside a finding, improving explainability.
- Computer-aided detection and triage algorithms — classification layers that convert model output into an actionable priority flag on the radiologist's worklist.
- Natural Language Processing (NLP) — used in newer report-drafting tools (e.g., Aidoc's First Read) to generate preliminary structured radiology report text from image findings.

Together these techniques form a computer-vision-centric pipeline, supplemented by NLP for the reporting stage — placing this application firmly within the Machine Learning / Deep Learning and Computer Vision branches of AI, with an emerging Generative AI component in report generation.

7. Working Mechanism

The system operates as a continuous pipeline that sits alongside the hospital's existing imaging infrastructure (PACS) rather than replacing it.

Working Mechanism of AI-Based Radiology Triage (e.g., Aidoc CARE Foundation Model)



Loop: model output continuously monitored & refined via aiOS governance and clinician feedback

Figure 1: Working mechanism of AI-based radiology triage, from image acquisition to radiologist sign-off.

- Input: a CT, X-ray, or MRI scan is captured by the imaging device and sent, as it normally would be, to the hospital's picture archiving system.
- Pre-processing: the AI pipeline intercepts a copy of the scan, normalizes it, and reduces noise so the image is in a consistent format for the model.
- Feature extraction: convolutional layers scan the image for learned visual patterns associated with specific conditions (e.g., the density signature of a bleed or a blocked vessel).
- Classification / triage: the foundation model scores the study against multiple possible acute findings at once and assigns a confidence-based flag.
- Worklist prioritization: flagged, time-critical studies are pushed to the top of the radiologist's queue, and in stroke-detection tools, an alert can be sent directly to a specialist's phone within minutes.
- Output: the radiologist reviews the flagged region with full imaging context and either confirms, adjusts, or overrides the AI suggestion before finalizing the report; newer tools additionally draft preliminary report text for the radiologist to edit.

Crucially, the AI functions as a decision-support layer, not an autonomous diagnostician — the final clinical decision and the report sign-off remain with the radiologist.

8. Data Sources and Processing

AI Technology Stack: Healthcare Diagnostic Imaging

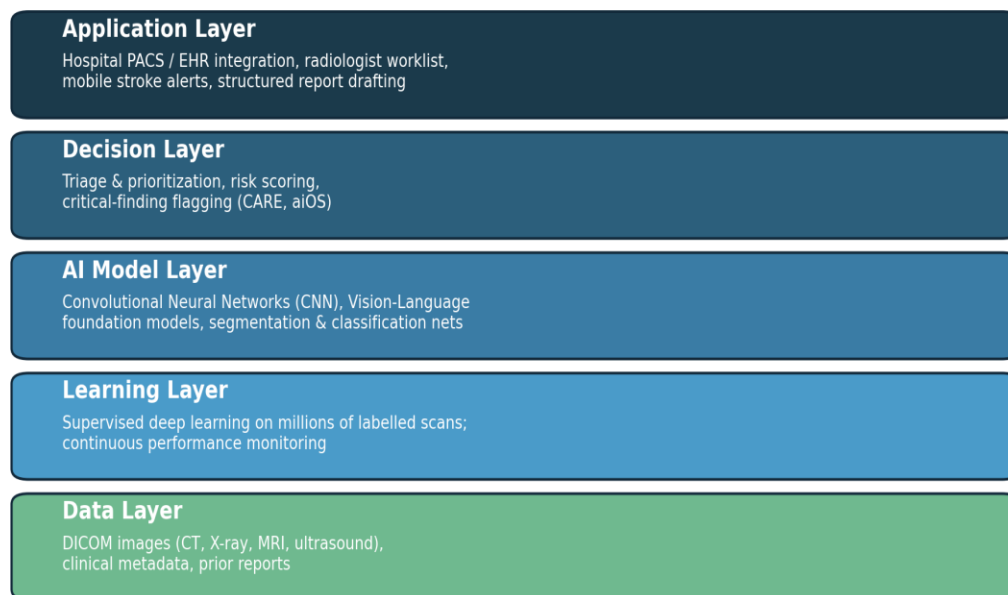


Figure 2: Layered technology stack underlying healthcare diagnostic-imaging AI.

The models are trained on very large, de-identified datasets of DICOM-format medical images (CT, X-ray, MRI, ultrasound) collected from multiple hospitals, each labelled by expert radiologists to indicate the presence or absence of specific findings. Aidoc's qXR-type products, for instance, have been trained on datasets exceeding nine million chest X-ray exams. During inference, the incoming patient scan is processed in the same normalized DICOM format, and the model's output is combined with clinical metadata (such as patient priority or exam type) before it reaches the worklist. Continuous performance monitoring and governance layers (Aidoc calls its version 'aiOS™') track live model accuracy after deployment and support periodic retraining, which is essential because imaging equipment, protocols, and patient populations vary across hospitals.

9. Real-World Implementation

Aidoc's platform is deployed at nearly 2,000 hospitals worldwide, including major U.S. health systems such as Sutter Health, WellSpan Health, Mercy, AdventHealth, Advocate Health, Mount Sinai, and Yale New Haven Health. In January 2026 the U.S. FDA cleared Aidoc's CARE™ platform as the first comprehensive foundation-model-powered clinical AI device — a single body-CT triage tool covering fourteen acute conditions such as aortic dissection, appendicitis, bowel obstruction, and spleen injury. In June 2026 the company's chest-X-ray report-drafting tool, First Read, received FDA Breakthrough Device Designation, its second such designation within a year. Comparable real-world deployments

include Viz.ai, whose stroke-detection algorithm sends real-time smartphone alerts to stroke teams and is active in more than 1,700 hospitals, and Qure.ai's qXR, deployed with 26 FDA-cleared indications and used, among others, by University Hospitals in Ohio to catch small pulmonary nodules that might otherwise be missed on routine chest X-rays.

10. Impact and Benefits

- **Accuracy:** in Aidoc's FDA-reviewed pivotal study covering eleven acute indications, the tool achieved a mean sensitivity of 97% (up to 98.5%) and a mean specificity of 98% (up to 99.7%).
- **Speed:** Qure.ai's qXR-Detect tool has been reported to cut chest X-ray reporting turnaround times by up to 40% at University Hospitals, Ohio.
- **Scale and adoption:** the number of FDA-cleared radiology-specific AI algorithms in the U.S. grew from around 151 in late 2021 to roughly 1,163 by March 2026, reflecting rapid mainstream clinical adoption.
- **Workflow efficiency:** automatic worklist re-ordering means the sickest patients are reviewed first rather than waiting in a first-come queue, directly supporting emergency-department throughput.
- **Reduced clinician fatigue:** by pre-screening every scan, the AI reduces the cognitive burden of manually triaging routine versus urgent studies, allowing radiologists to focus expertise on complex cases.

11. Challenges and Limitations

- **Explainability:** many deep learning models historically functioned as a 'black box', giving little insight into why a finding was flagged — a documented barrier to clinician trust and adoption, which newer reasoning-based models are only beginning to address.
- **Data privacy and security:** training and operating on large volumes of patient imaging data raises strict compliance requirements (e.g., HIPAA) and the risk of data breaches or misuse.
- **Integration complexity:** connecting an AI triage layer to a hospital's existing PACS/EHR infrastructure without disrupting clinical workflows requires significant IT effort and radiologist retraining.
- **Generalization across sites:** a model trained on one hospital's scanners and patient population can perform less reliably at another site with different equipment or demographics, which is why continuous post-deployment monitoring is necessary.
- **Accountability and over-reliance:** because the AI only supports, and does not replace, the radiologist, there is an ongoing ethical and legal question of how responsibility is shared if an AI-flagged (or AI-missed) finding contributes to a diagnostic error, and a related risk of clinicians over-trusting automated flags.

- Equity of access: high-income health systems have adopted these tools far faster than smaller or resource-constrained hospitals, risking a widening gap in diagnostic quality between well-funded and under-funded facilities.

12. Future Scope and Emerging Trends

- Foundation models covering all CT and X-ray workflows: Aidoc has indicated its CARE™ roadmap is expected to expand across the full range of CT and X-ray studies over the next 18 months, moving from single-condition tools toward one comprehensive triage layer.
- AI-drafted radiology reports: tools such as First Read point toward AI increasingly assisting not just detection but the actual writing of preliminary report text, with the radiologist reviewing and finalizing it.
- Explainable, reasoning-based models: vision-language models that narrate their reasoning step-by-step (rather than returning only a confidence score) are an emerging response to the explainability gap.
- Expansion to underserved regions: AI triage is increasingly positioned to help offset radiologist shortages in low-resource settings and rural hospitals, extending diagnostic capability where specialists are scarce.
- Multimodal AI: future systems are expected to combine imaging with structured clinical data, lab results, and patient history for more holistic, context-aware risk scoring rather than image analysis alone.

13. Conclusion

AI-powered radiology triage represents one of the most mature and clinically validated applications of artificial intelligence in healthcare today. By using convolutional neural networks and, increasingly, foundation models to continuously scan, classify, and prioritize medical images, platforms such as Aidoc's CARE™, Viz.ai, and Qure.ai's qXR are measurably reducing diagnostic delay for time-critical conditions and easing the workload on radiologists. The technology's rapid FDA clearance growth and deployment across thousands of hospitals demonstrate genuine, scaled real-world impact rather than experimental promise alone. At the same time, unresolved challenges around explainability, data privacy, cross-site generalization, and equitable access mean the technology must continue to be deployed as a decision-support tool under clinician oversight, not as an autonomous replacement for medical expertise. As foundation models, explainable AI, and multimodal reasoning mature, this application is well positioned to become a standard, trusted layer of hospital diagnostic infrastructure worldwide.

14. References

1. Pinggy, "AI Medical Imaging in 2026: Best Radiology AI Tools, FDA Clearances, and Diagnostic Accuracy," pinggy.io, 2026.
2. buildmvpfast.com, "FDA-Cleared AI Radiology Tools Hospitals Actually Use," 2026.
3. Aidoc, "Aidoc Receives FDA Breakthrough Device Designation for AI That Drafts Radiology Reports," PR Newswire / aidoc.com, June 25, 2026.
4. buildmvpfast.com, "FDA-Cleared AI Radiology Tools Hospitals Actually Use," Mar. 22, 2026.
5. RadiologyBusiness.com, "FDA has now cleared 700 AI healthcare algorithms, more than 76% in radiology," 2025.

Rubrics for Evaluation

S.No	Criteria	Weightage (%)	Excellent (100–90%)	Good (89–75%)	Satisfactory (74–60%)	Improvement
1.	Identification of Latest AI Application	10%	Selects a highly relevant, emerging AI application with strong justification.	Selects a recent and relevant AI application.	Selects a suitable application with limited justification.	Application is outdated or irrelevant.
2.	Problem Analysis and Domain Understanding	15%	Clearly explains the real-world problem, domain context, and need for AI with critical analysis.	Good explanation with adequate analysis.	Basic explanation with limited analysis.	Problem statement is unclear or incomplete.
3.	AI Technologies and Working Mechanism	20%	Accurately explains AI techniques, algorithms, architecture, and workflow with technical depth.	Good explanation with minor omissions.	Covers only basic AI concepts.	AI technologies are poorly explained or incorrect.
4.	Research Quality and Literature Review	10%	Uses recent, credible sources with proper citations and excellent synthesis.	Good quality references with appropriate citations.	Limited references with basic discussion.	Few or unreliable references; poor citation.

5.	Real-World Implementation and Case Analysis	10%	Thoroughly analyzes industrial implementation with appropriate examples.	Good implementation analysis.	Basic implementation discussion.	Minimal or no real-world analysis.
6.	Impact, Challenges, and Future Scope	15%	Critically evaluates benefits, limitations, ethical issues, and future trends.	Good evaluation with minor gaps.	Basic discussion of impact and future scope.	Limited evaluation or missing sections.
7.	Originality and Critical Thinking	10%	Demonstrates independent research, innovation, and critical evaluation.	Good analytical thinking.	Some originality with limited analysis.	Mostly descriptive or copied content.
8.	Organization, Presentation, and Documentation	10%	Exceptionally organized, professional presentation, error-free, and properly referenced.	Well organized with minor issues.	Adequately organized with some errors.	Poor organization and documentation.